Boston House Prices: Multilayer Perceptron Regression Report

Dataset

The supplied file contains 506 rows and 14 whitespace-separated numeric columns. Columns 1-13 form the feature matrix and column 14, MEDV, is the regression target. MEDV is the median value of owner-occupied homes in thousands of dollars.

No.	Variable	Definition
1	CRIM	Per-capita crime rate by town
2	ZN	Proportion of residential land zoned for lots over 25,000 sq. ft.
3	INDUS	Proportion of non-retail business acres per town
4	CHAS	Charles River indicator: 1 when the tract bounds the river, otherwise 0
5	NOX	Nitric oxides concentration, parts per 10 million
6	RM	Average number of rooms per dwelling
7	AGE	Proportion of owner-occupied units built before 1940
8	DIS	Weighted distance to five Boston employment centres
9	RAD	Index of accessibility to radial highways
10	TAX	Full-value property-tax rate per $10,000
11	PTRATIO	Pupil-teacher ratio by town
12	B	1000(Bk - 0.63)^2, where Bk is the proportion of Black residents by town
13	LSTAT	Percentage of the population with lower status
14	MEDV	Median value of owner-occupied homes, in $1,000s

The executed program reported MEDV mean 22.53, standard deviation 9.20, minimum 5.00, and maximum 50.00. It also reported feature values on substantially different scales, including NOX maximum 0.87 and TAX maximum 711.00.

Steps 1-5: Two-hidden-layer MLP

The PyTorch model contains fully connected layers with architecture 13-64-32-1. It applies ReLU after each hidden layer, has a linear single-unit output layer, uses Adam with learning rate 0.001, and trains with mean squared error loss.

The evaluation used shuffled 10-fold cross-validation, 150 epochs, batch size 16, and random state 7. The standardized model used a Pipeline with StandardScaler before the estimator, so the scaler is fit independently within each training split.

Setup	Mean CV MSE	SD of fold MSE	Per-fold MSE
13-64-32-1 MLP without scaling	18.82	7.37	Not retained for this configuration
13-64-32-1 MLP with StandardScaler	12.17	5.63	7.11, 25.09, 5.69, 17.53, 15.83, 12.91, 11.49, 8.70, 9.71, 7.62

The standardized pipeline's mean MSE was 12.17 in the fresh run, below the assignment's stated reference value of approximately 20. The unscaled result was 18.82, so this execution associated scaling with a lower mean cross-validation MSE by 6.65.

Step 6: One hidden layer with ReLU

OneLayerNet is a NumPy implementation of one fully connected ReLU hidden layer and one output unit. The fit method derives its input width from the supplied feature matrix and accepts hidden_units as a parameter; the model therefore supports any input width and requested number of hidden units while retaining one output. The implementation uses hand-derived backpropagation, He initialization, and a NumPy Adam implementation.

The width experiment used the same standardized 10-fold evaluation approach, 150 epochs, batch size 16, and learning rate 0.001.

Hidden units	Mean CV MSE	SD of fold MSE
1	420.92	59.32
2	421.86	59.32
4	26.90	12.74
8	21.59	8.52
16	18.19	8.57
32	13.73	8.81
64	13.17	8.91
128	12.85	8.93
256	12.00	7.12

In this run, widths of 16 or more had mean MSE below 20, whereas 1 and 2 units produced mean MSE above 420. The test therefore identified a usable width at 16 units without a fine-grained search; 32-256 units produced lower MSEs in the same run.

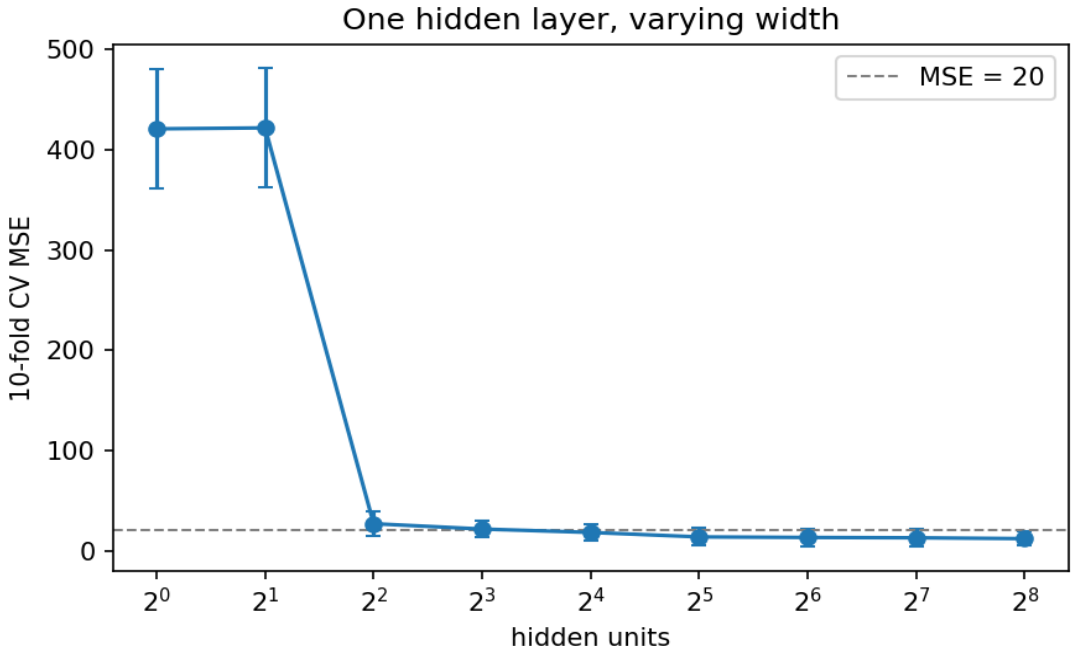


Figure 1. Ten-fold cross-validation MSE by one-hidden-layer width.

Step 7: Arbitrary-depth network

DeepNet accepts a tuple of hidden-layer widths, constructs one weight matrix and bias vector per layer, applies ReLU to all hidden layers, and leaves the output layer linear. Its forward pass retains activations and pre-activations, and its backward pass traverses layers in reverse order.

The program checked the one-layer special case before the depth experiment: DeepNet(hidden_layers=(16,)) and OneLayerNet(hidden_units=16) produced a maximum prediction difference of 0.0 after five epochs on the standardized full dataset.

For the depth experiment, each hidden layer had 32 units. The program evaluated 1, 2, 3, 4, 5, 6, and 8 hidden layers across five folds and three seeds, for 15 runs per depth. Each run used 200 epochs, batch size 16, learning rate 0.001, and training-fold standardization.

Hidden layers	Parameters	Epoch to val MSE <= 25	Epoch to val MSE <= 15	Best val MSE	Final val MSE	Final train MSE
1	481	38 (15/15)	51 (9/15)	13.86	13.87	8.21
2	1,537	11 (15/15)	16 (10/15)	12.63	12.74	3.93
3	2,593	7 (15/15)	18 (14/15)	11.56	11.78	1.71
4	3,649	6 (15/15)	16 (14/15)	12.70	13.42	1.11
5	4,705	7 (15/15)	13 (11/15)	13.26	13.50	0.76
6	5,761	6 (15/15)	12 (12/15)	13.28	13.92	0.65
8	7,873	5 (15/15)	10 (10/15)	14.13	15.70	0.57

In this experiment, the three-layer network attained the smallest best mean validation MSE (11.56). The one-layer model first reached validation MSE 25 at epoch 38, while the models with 3, 4, 5, 6, and 8 layers reached that threshold at median epochs 7, 6, 7, 6, and 5 respectively. These are observations from this fixed architecture and evaluation design rather than a general claim about all neural-network depths.

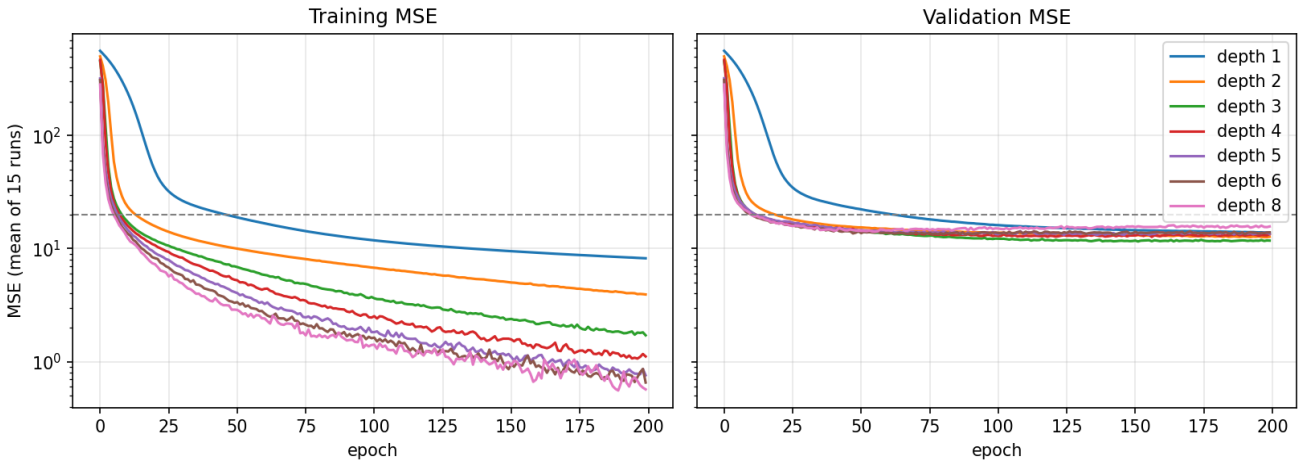


Figure 2. Mean training and validation MSE across 15 runs by network depth.

Conclusion

The completed program satisfies the requested dataset loading, feature/target split, MLP construction, fold-local standardization, k-fold MSE evaluation, configurable one-hidden-layer implementation, and arbitrary-depth implementation. The strongest mean validation result reported by the executed experiments was 11.56 for the three-layer 32-unit configuration; the standardized two-hidden-layer PyTorch model reported 12.17 mean MSE.